

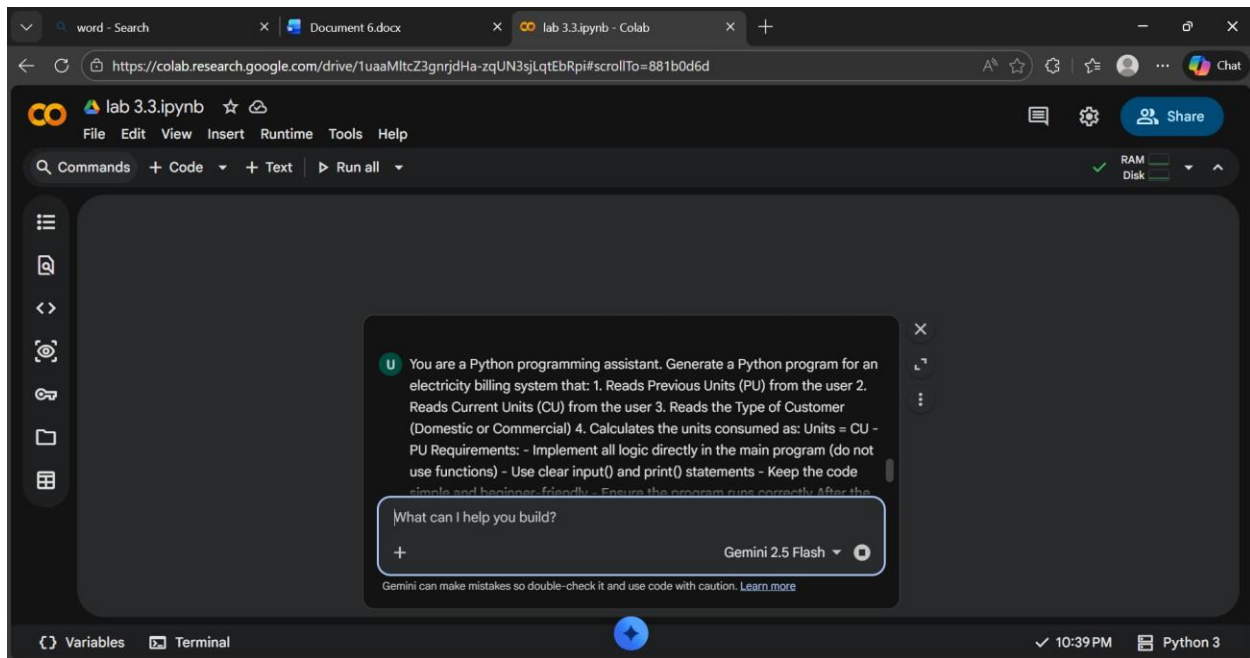
Assignment 3.3 AI ASSISTED CODING

Htno:2303a51983

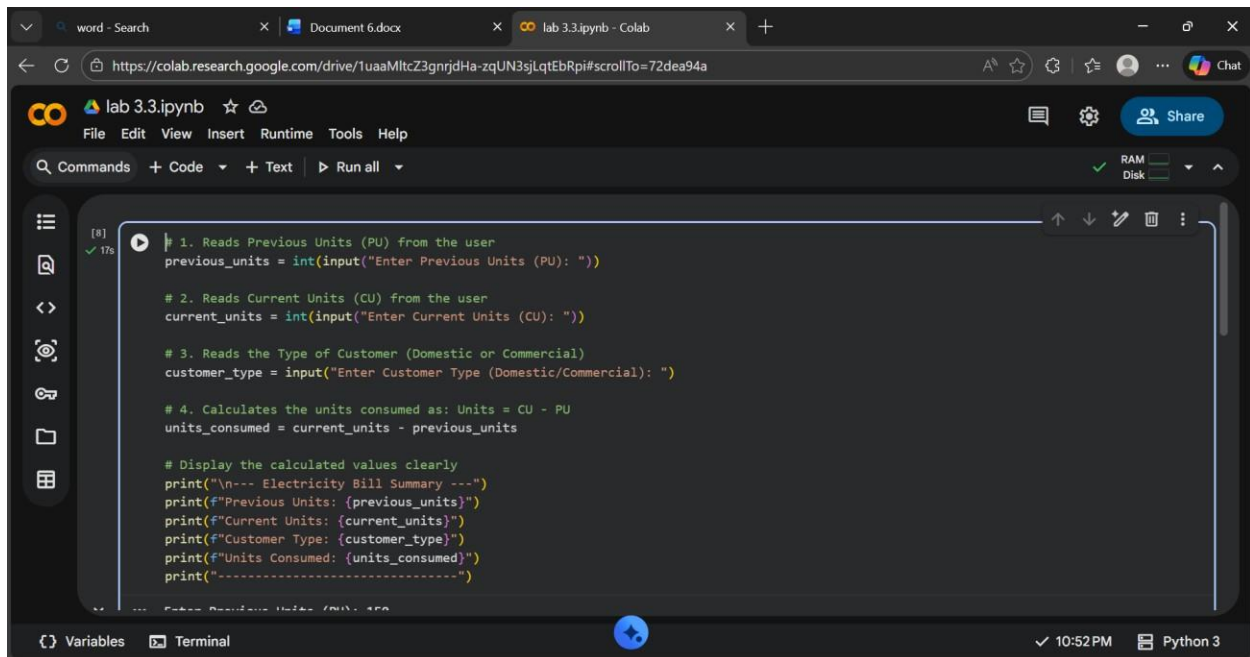
Btno:06

Task 1: AI-Generated Logic for Reading Consumer Details .

Prompt:



Code:



The screenshot shows a Google Colab notebook interface. The top bar includes the Colab logo, the file name 'lab 3.3.ipynb', and a 'Share' button. Below the top bar is a menu bar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar and a 'Run all' button are also present. The main code area contains the following Python code:

```
[8] ✓ 17s
1. Reads Previous Units (PU) from the user
previous_units = int(input("Enter Previous Units (PU): "))

# 2. Reads Current Units (CU) from the user
current_units = int(input("Enter Current Units (CU): "))

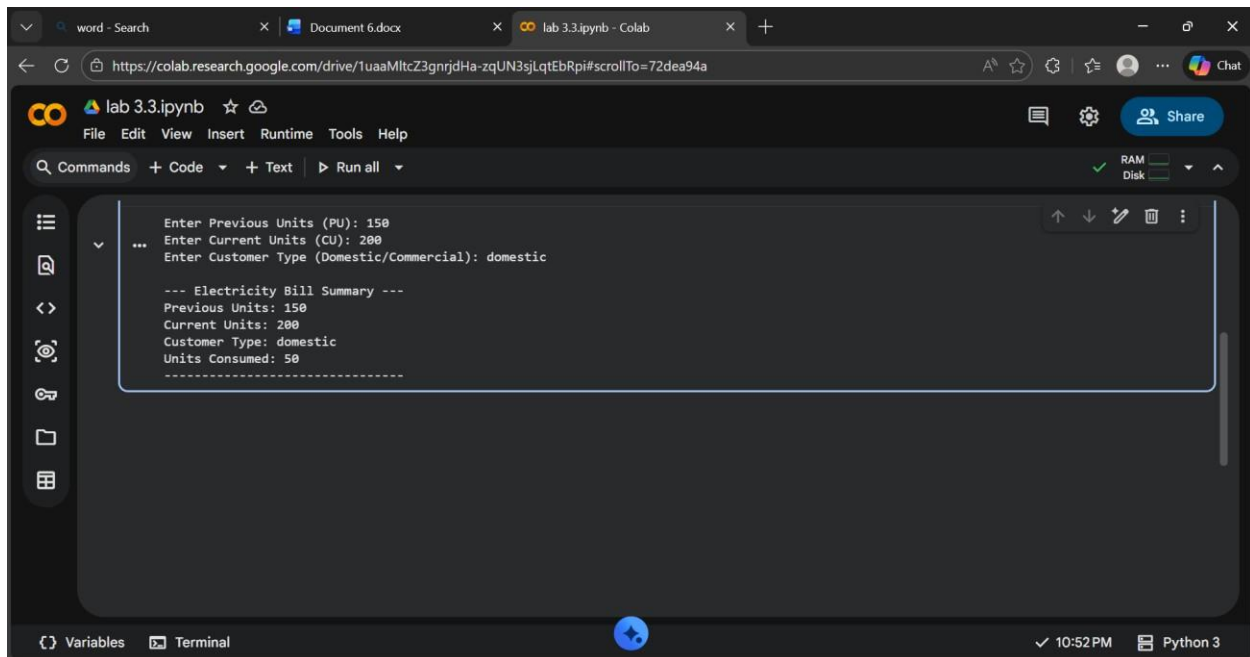
# 3. Reads the Type of Customer (Domestic or Commercial)
customer_type = input("Enter Customer Type (Domestic/Commercial): ")

# 4. Calculates the units consumed as: Units = CU - PU
units_consumed = current_units - previous_units

# Display the calculated values clearly
print("\n--- Electricity Bill Summary ---")
print(f"Previous Units: {previous_units}")
print(f"Current Units: {current_units}")
print(f"Customer Type: {customer_type}")
print(f"Units Consumed: {units_consumed}")
print("-----")
```

The bottom of the interface shows a 'Variables' tab and a 'Terminal' tab. The status bar at the bottom indicates '10:52 PM' and 'Python 3'.

Output:



The screenshot shows the same Google Colab notebook interface, but now displaying the output of the code. The output is as follows:

```
Enter Previous Units (PU): 150
Enter Current Units (CU): 200
Enter Customer Type (Domestic/Commercial): domestic

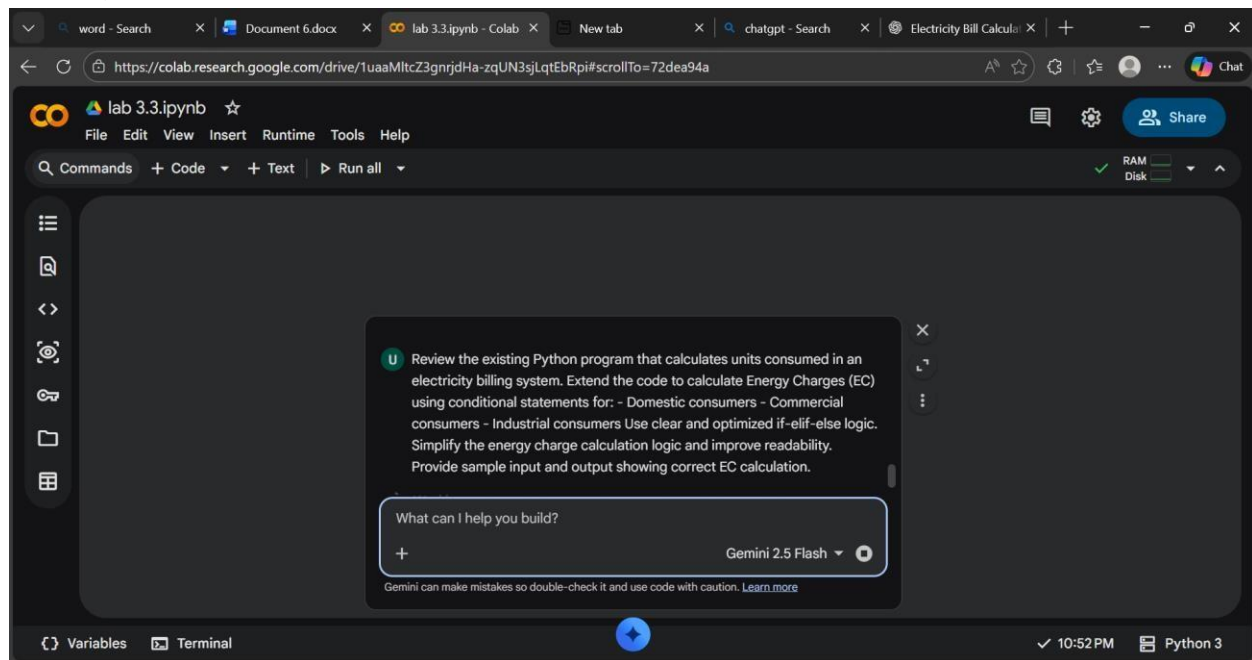
--- Electricity Bill Summary ---
Previous Units: 150
Current Units: 200
Customer Type: domestic
Units Consumed: 50
-----
```

The bottom of the interface shows the 'Variables' and 'Terminal' tabs. The status bar at the bottom indicates '10:52 PM' and 'Python 3'.

Explanation: The program reads previous units, current units, and customer type from the user. It calculates **units consumed** by subtracting previous units from current units ($200 - 150 = 50$). Finally, it displays a clear electricity bill summary with all entered details and the calculated consumption.

Task 2: Energy Charges Calculation Based on Units Consumed.

Prompt:



Code:

The image displays two sequential screenshots of a Google Colab notebook titled 'lab 3.3.ipynb'. The interface includes a top toolbar with 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help' menus, and a 'Share' button. The left sidebar contains icons for file management and a 'Variables' panel. The main area shows a Python code editor with a dark theme.

First Screenshot: The code defines variables for previous units, current units, and customer type. It calculates units consumed and sets default tariff rates.

```
[10] ✓ 10s
# 1. Reads Previous Units (PU) from the user
previous_units = int(input("Enter Previous Units (PU): "))

# 2. Reads Current Units (CU) from the user
current_units = int(input("Enter Current Units (CU): "))

# 3. Reads the Type of Customer (Domestic or Commercial or Industrial)
customer_type = input("Enter Customer Type (Domestic/Commercial/Industrial): ").strip().lower()

# 4. Calculates the units consumed as: Units = CU - PU
units_consumed = current_units - previous_units

# Define tariff rates (example rates)
domestic_rate_per_unit = 5.0
commercial_rate_per_unit = 8.0
industrial_rate_per_unit = 10.0

energy_charges = 0.0

# Calculate Energy Charges (EC) using conditional statements
if customer_type == 'domestic':
    energy_charges = units_consumed * domestic_rate_per_unit
```

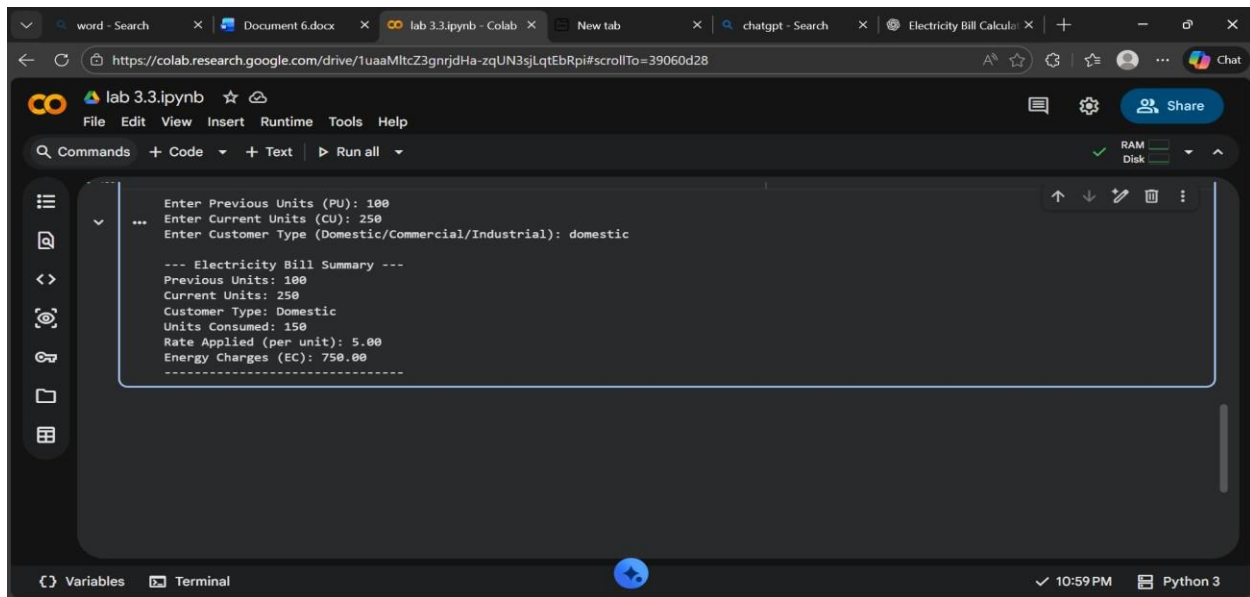
Second Screenshot: The code is updated with conditional logic for different customer types and formatted output.

```
[10] ✓ 10s
if customer_type == 'domestic':
    energy_charges = units_consumed * domestic_rate_per_unit
    rate_applied = domestic_rate_per_unit
elif customer_type == 'commercial':
    energy_charges = units_consumed * commercial_rate_per_unit
    rate_applied = commercial_rate_per_unit
elif customer_type == 'industrial':
    energy_charges = units_consumed * industrial_rate_per_unit
    rate_applied = industrial_rate_per_unit
else:
    print("Invalid customer type entered. Energy charges cannot be calculated.")
    rate_applied = 0.0 # Assign a default or indicate no rate

# Display the calculated values clearly
print("\n--- Electricity Bill Summary ---")
print(f"Previous Units: {previous_units}")
print(f"Current Units: {current_units}")
print(f"Customer Type: {customer_type.capitalize()}")
print(f"Units Consumed: {units_consumed}")

if energy_charges > 0 or customer_type in ['domestic', 'commercial', 'industrial']:
    print(f"Rate Applied (per unit): {rate_applied:.2f}")
```

Output:



```
Enter Previous Units (PU): 100
Enter Current Units (CU): 250
Enter Customer Type (Domestic/Commercial/Industrial): domestic

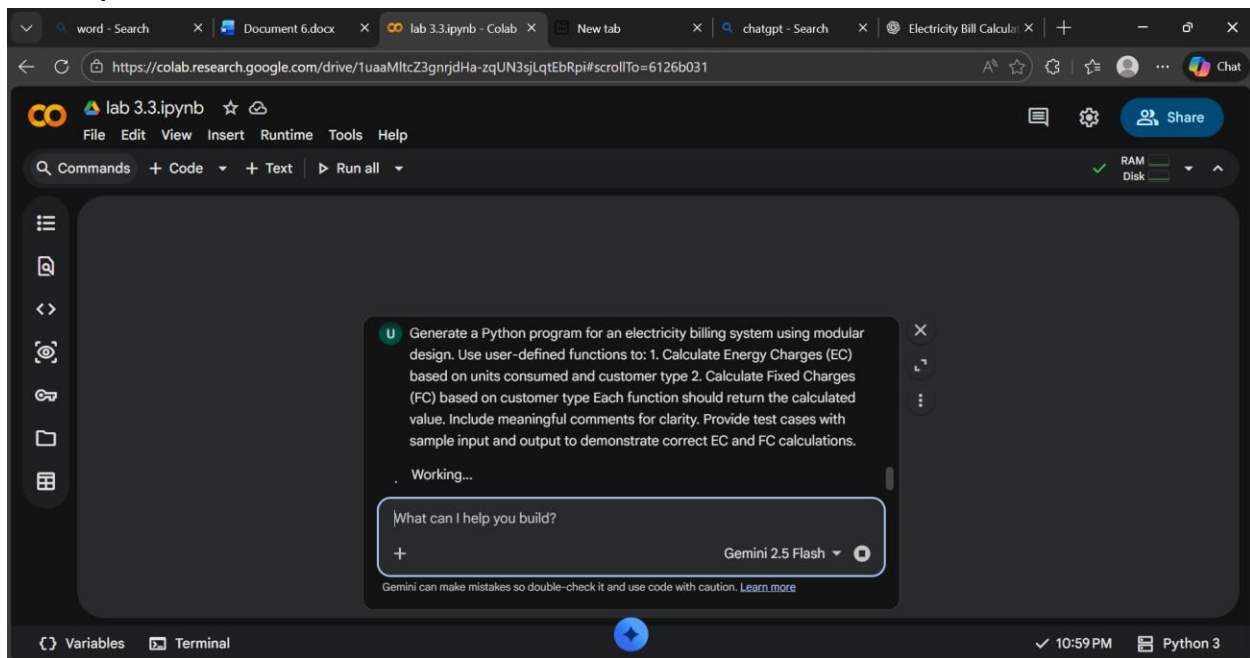
--- Electricity Bill Summary ---
Previous Units: 100
Current Units: 250
Customer Type: Domestic
Units Consumed: 150
Rate Applied (per unit): 5.00
Energy Charges (EC): 750.00
-----
```

Explanation: The extended program calculates **Energy Charges** based on units consumed and customer type using conditional statements.

Optimized if-elif-else logic improves readability and makes the billing rules easy to understand.

Task 3: Modular Design Using AI Assistance (Using Functions).

Prompt:



```
U Generate a Python program for an electricity billing system using modular design. Use user-defined functions to: 1. Calculate Energy Charges (EC) based on units consumed and customer type 2. Calculate Fixed Charges (FC) based on customer type Each function should return the calculated value. Include meaningful comments for clarity. Provide test cases with sample input and output to demonstrate correct EC and FC calculations.

Working...

What can I help you build?
+ Gemini 2.5 Flash
```

Code:

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lab 3.3.ipynb | File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
[11] ✓ 12s
# --- Define Tariff Rates and Fixed Charges ---
# Rates per unit for different customer types
DOMESTIC_RATE_PER_UNIT = 5.0
COMMERCIAL_RATE_PER_UNIT = 8.0
INDUSTRIAL_RATE_PER_UNIT = 10.0

# Fixed charges for different customer types
DOMESTIC_FIXED_CHARGE = 50.0
COMMERCIAL_FIXED_CHARGE = 150.0
INDUSTRIAL_FIXED_CHARGE = 300.0

# --- Function to Calculate Energy Charges (EC) ---
def calculate_energy_charges(units_consumed, customer_type):
    """
    Calculates Energy Charges based on units consumed and customer type.
    Returns the calculated energy charges.
    """
    energy_charge = 0.0
    rate_applied = 0.0

    if customer_type == 'domestic':
        energy_charge = units_consumed * DOMESTIC_RATE_PER_UNIT
```

To undo cell deletion use Ctrl+M Z or the Undo option in the Edit menu

✓ 11:02 PM Python 3

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lab 3.3.ipynb | File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
[11] ✓ 12s
if customer_type == 'domestic':
    energy_charge = units_consumed * DOMESTIC_RATE_PER_UNIT
    rate_applied = DOMESTIC_RATE_PER_UNIT
elif customer_type == 'commercial':
    energy_charge = units_consumed * COMMERCIAL_RATE_PER_UNIT
    rate_applied = COMMERCIAL_RATE_PER_UNIT
elif customer_type == 'industrial':
    energy_charge = units_consumed * INDUSTRIAL_RATE_PER_UNIT
    rate_applied = INDUSTRIAL_RATE_PER_UNIT
else:
    print(f"Warning: Unknown customer type '{customer_type}'. Energy charges set to 0.")

return energy_charge, rate_applied

# --- Function to Calculate Fixed Charges (FC) ---
def calculate_fixed_charges(customer_type):
    """
    Calculates Fixed Charges based on customer type.
    Returns the calculated fixed charges.
    """
    fixed_charge = 0.0
```

() Variables | Terminal

✓ 11:02 PM Python 3

```
[11] ✓ 12s  
def calculate_fixed_charges(customer_type):  
    if customer_type == 'domestic':  
        fixed_charge = DOMESTIC_FIXED_CHARGE  
    elif customer_type == 'commercial':  
        fixed_charge = COMMERCIAL_FIXED_CHARGE  
    elif customer_type == 'industrial':  
        fixed_charge = INDUSTRIAL_FIXED_CHARGE  
    else:  
        # Warning already handled in energy charges, or can be added here too  
        pass  
    return fixed_charge  
  
# --- Main Program Logic ---  
print("--- Electricity Billing System ---")  
  
# 1. Read Previous Units (PU) from the user  
previous_units = int(input("Enter Previous Units (PU): "))  
  
# 2. Read Current Units (CU) from the user  
current_units = int(input("Enter Current Units (CU): "))  
  
# 3. Read the Type of Customer (Domestic, Commercial, or Industrial)  
customer_type = input("Enter Customer Type (Domestic/Commercial/Industrial): ").strip().lower()  
  
# Calculate Fixed Charges using the function  
fixed_charges = calculate_fixed_charges(customer_type)  
  
# Calculate Total Bill  
total_bill = energy_charges + fixed_charges  
  
# --- Display Bill Summary ---  
print("\n--- Bill Summary ---")  
print(f"Previous Units: {previous_units}")  
print(f"Current Units: {current_units}")  
print(f"Customer Type: {customer_type.capitalize()}")  
print(f"Units Consumed: {units_consumed}")  
  
if rate_applied > 0:  
    print(f"Rate Applied (per unit): {rate_applied:.2f}")  
  
print(f"Energy Charges (EC): {energy_charges:.2f}")  
print(f"Fixed Charges (FC): {fixed_charges:.2f}")  
print(f"Total Bill: {total_bill:.2f}")  
print("-----")
```

Output:

The screenshot shows a Google Colab notebook titled 'lab 3.3.ipynb'. The code cell contains a Python program for an electricity billing system. The program prompts the user for 'Previous Units (PU)', 'Current Units (CU)', and 'Customer Type (Domestic/Commercial/Industrial)'. It then calculates and displays a 'Bill Summary' including 'Previous Units', 'Current Units', 'Units Consumed', 'Rate Applied (per unit)', 'Energy Charges (EC)', 'Fixed Charges (FC)', and 'Total Bill'.

```
--- Electricity Billing System ---
Enter Previous Units (PU): 100
Enter Current Units (CU): 250
Enter Customer Type (Domestic/Commercial/Industrial): domestic

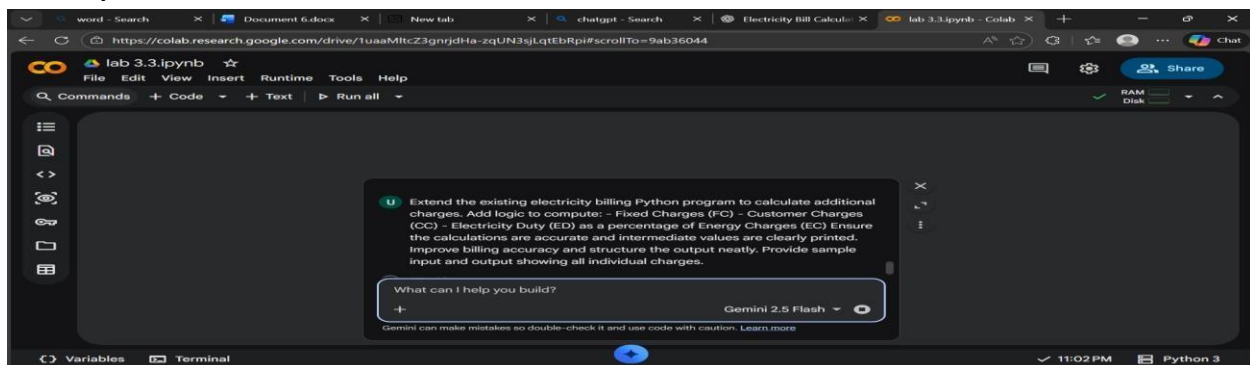
--- Bill Summary ---
Previous Units: 100
Current Units: 250
Customer Type: Domestic
Units Consumed: 150
Rate Applied (per unit): 5.00
Energy Charges (EC): 750.00
Fixed Charges (FC): 50.00
Total Bill: 800.00
-----
```

Explanation: This program uses **functions** to make the billing logic reusable for multiple consumers.

Separating Energy Charges and Fixed Charges into functions improves clarity, modularity, and easy maintenance of the code.

Task 4: Calculation of Additional Charges.

Prompt:



Code:

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lab 3.3.ipynb ☆ Share

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RAM Disk

```
[13] ✓ 8s
--2# --- Define Tariff Rates and Fixed Charges ---
+# --- Define Tariff Rates and Fixed Charges ---
# Rates per unit for different customer types
DOMESTIC_RATE_PER_UNIT = 5.0
COMMERCIAL_RATE_PER_UNIT = 8.0
INDUSTRIAL_RATE_PER_UNIT = 10.0

# Fixed charges for different customer types
DOMESTIC_FIXED_CHARGE = 50.0
COMMERCIAL_FIXED_CHARGE = 150.0
INDUSTRIAL_FIXED_CHARGE = 300.0

# Customer Charges (assuming a flat rate for simplicity for all types)
CUSTOMER_CHARGE = 20.0

# Electricity Duty (as a percentage of Energy Charges)
ELECTRICITY_DUTY_PERCENTAGE = 0.05 # 5%

--# --- Function to Calculate Energy Charges (EC) ---
+# --- Function to Calculate Energy Charges (EC) ---
def calculate_energy_charges(units_consumed, customer_type):
    """
```

Variables Terminal

✓ 11:07 PM Python 3

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lab 3.3.ipynb ☆ Share

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RAM Disk

```
[13] ✓ 8s
Returns the calculated energy charges and the rate applied.
"""
energy_charge = 0.0
rate_applied = 0.0

if customer_type == 'domestic':
    energy_charge = units_consumed * DOMESTIC_RATE_PER_UNIT
    rate_applied = DOMESTIC_RATE_PER_UNIT
elif customer_type == 'commercial':
    energy_charge = units_consumed * COMMERCIAL_RATE_PER_UNIT
    rate_applied = COMMERCIAL_RATE_PER_UNIT
elif customer_type == 'industrial':
    energy_charge = units_consumed * INDUSTRIAL_RATE_PER_UNIT
    rate_applied = INDUSTRIAL_RATE_PER_UNIT
else:
    print(f"Warning: Unknown customer type '{customer_type}'. Energy charges set to 0.")

return energy_charge, rate_applied

--# --- Function to Calculate Fixed Charges (FC) ---
+# --- Function to Calculate Fixed Charges (FC) ---
def calculate_fixed_charges(customer_type):
    """
```

Variables Terminal

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lab 3.3.ipynb

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RAM Disk

[13] 8s

```
Calculates Fixed Charges based on customer type.
Returns the calculated fixed charges.
"""
fixed_charge = 0.0
if customer_type == 'domestic':
    fixed_charge = DOMESTIC_FIXED_CHARGE
elif customer_type == 'commercial':
    fixed_charge = COMMERCIAL_FIXED_CHARGE
elif customer_type == 'industrial':
    fixed_charge = INDUSTRIAL_FIXED_CHARGE
return fixed_charge

--# --- Function to Calculate Customer Charges (CC) ---
+# --- Function to Calculate Customer Charges (CC) ---
def calculate_customer_charges(customer_type):
    """
    Calculates Customer Charges.
    For simplicity, this example uses a flat rate for all customer types.
    """
    # This function could be expanded to have type-dependent customer charges if needed.
    return CUSTOMER_CHARGE
```

Variables Terminal

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lab 3.3.ipynb

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RAM Disk

[13] 8s

```
--# --- Function to Calculate Electricity Duty (ED) ---
+# --- Function to Calculate Electricity Duty (ED) ---
def calculate_electricity_duty(energy_charges):
    """
    Calculates Electricity Duty as a percentage of Energy Charges.
    """
    return energy_charges * ELECTRICITY_DUTY_PERCENTAGE

--# --- Main Program Logic ---
+# --- Main Program Logic ---
print("--- Electricity Billing System (Extended) ---")

# 1. Read Previous Units (PU) from the user
previous_units = int(input("Enter Previous Units (PU): "))

# 2. Read Current Units (CU) from the user
current_units = int(input("Enter Current Units (CU): "))

# 3. Read the Type of Customer (Domestic, Commercial, or Industrial)
customer_type = input("Enter Customer Type (Domestic/Commercial/Industrial): ").strip().lower()

# Calculate Units Consumed
```

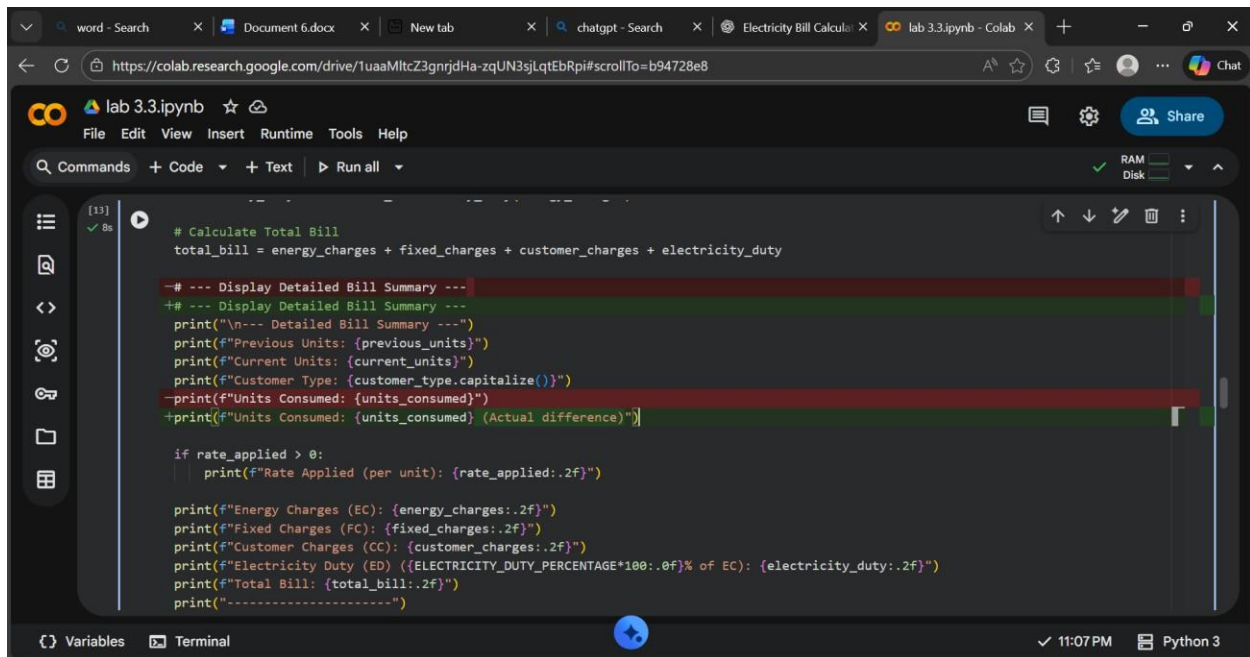
Variables Terminal

11:07 PM Python 3

Snipping Tool

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Markup and share



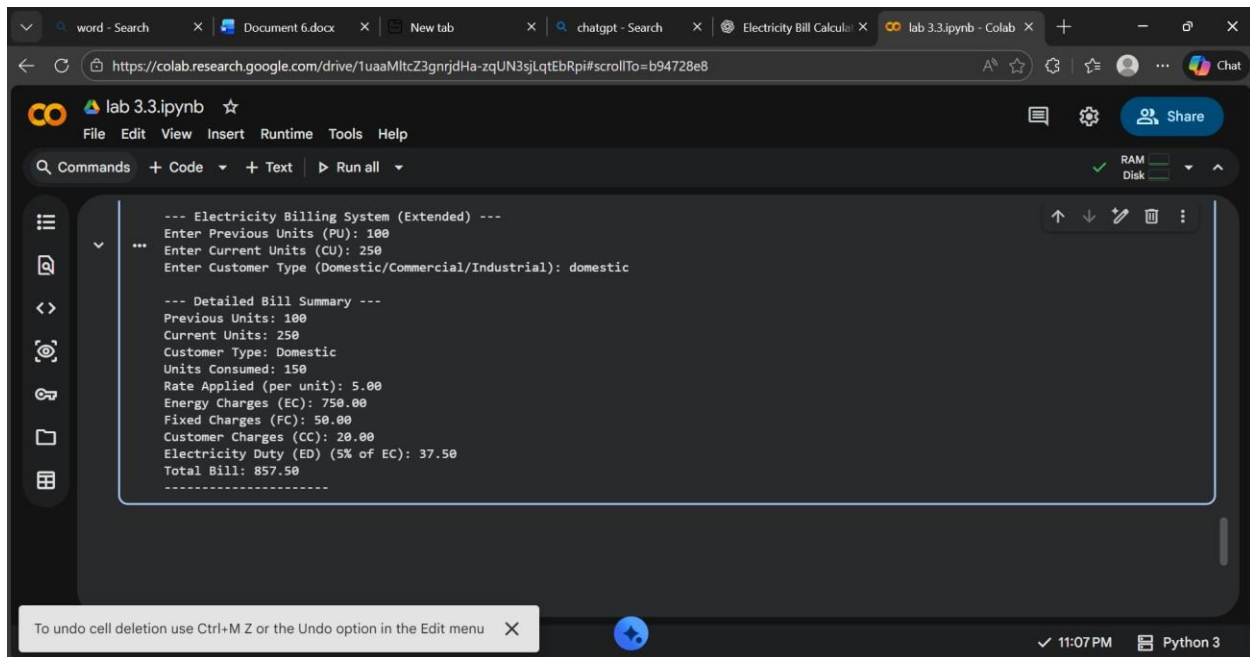
```
[13] ✓ 8s
# Calculate Total Bill
total_bill = energy_charges + fixed_charges + customer_charges + electricity_duty

--# --- Display Detailed Bill Summary ---
+# --- Display Detailed Bill Summary ---
print("\n--- Detailed Bill Summary ---")
print(f"Previous Units: {previous_units}")
print(f"Current Units: {current_units}")
print(f"Customer Type: {customer_type.capitalize()}")
print(f"Units Consumed: {units_consumed}")
+print(f"Units Consumed: {units_consumed} (Actual difference)")

if rate_applied > 0:
    print(f"Rate Applied (per unit): {rate_applied:.2f}")

print(f"Energy Charges (EC): {energy_charges:.2f}")
print(f"Fixed Charges (FC): {fixed_charges:.2f}")
print(f"Customer Charges (CC): {customer_charges:.2f}")
print(f"Electricity Duty (ED) ({ELECTRICITY_DUTY_PERCENTAGE*100:.0f}% of EC): {electricity_duty:.2f}")
print(f"Total Bill: {total_bill:.2f}")
print("-----")
```

Output:



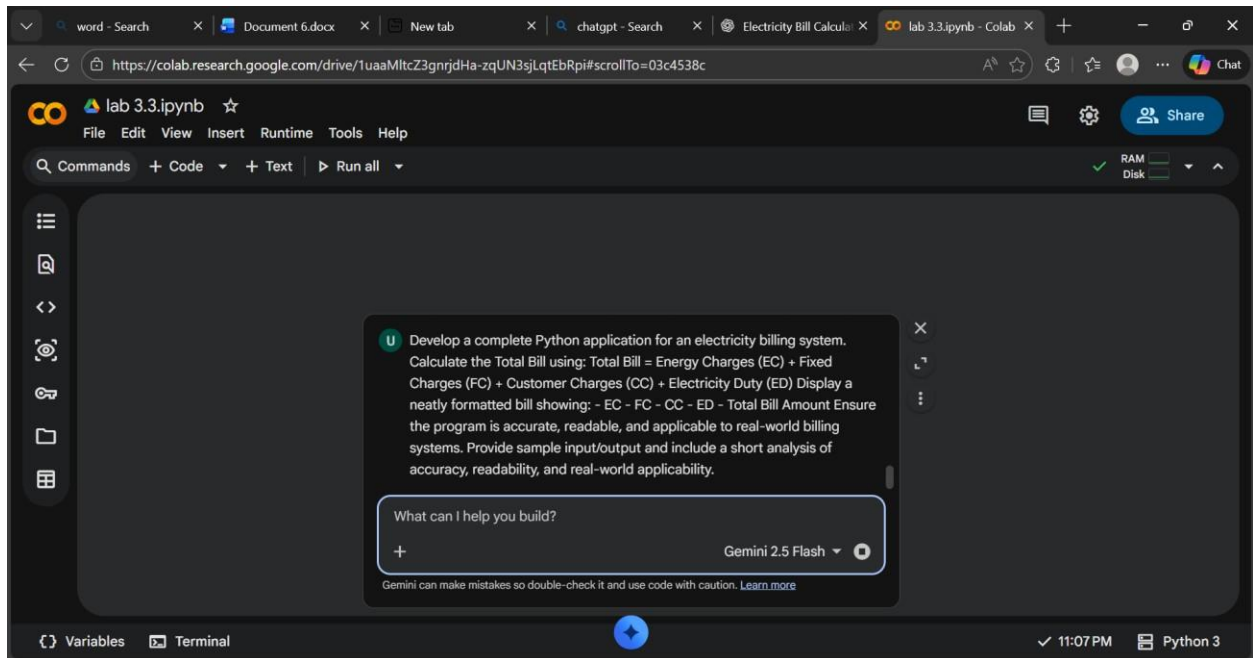
```
--- Electricity Billing System (Extended) ---
Enter Previous Units (PU): 100
Enter Current Units (CU): 250
Enter Customer Type (Domestic/Commercial/Industrial): domestic

--- Detailed Bill Summary ---
Previous Units: 100
Current Units: 250
Customer Type: Domestic
Units Consumed: 150
Rate Applied (per unit): 5.00
Energy Charges (EC): 750.00
Fixed Charges (FC): 50.00
Customer Charges (CC): 20.00
Electricity Duty (ED) (5% of EC): 37.50
Total Bill: 857.50
-----
```

Explanation: The program is enhanced to include **additional billing components** like fixed charges, customer charges, and electricity duty. Electricity Duty is calculated as a **percentage of Energy Charges**, and all values are displayed clearly for verification.

Task 5: Final Bill Generation and Output Analysis.

Prompt:



Code:

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lab 3.3.ipynb

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RAM Disk

```
[15] import pandas as pd

# 1. Define global constants for domestic, commercial, and industrial rates per unit
DOMESTIC_RATE_PER_UNIT = 5.0
COMMERCIAL_RATE_PER_UNIT = 8.0
INDUSTRIAL_RATE_PER_UNIT = 10.0

# 2. Define global constants for domestic, commercial, and industrial fixed charges
DOMESTIC_FIXED_CHARGE = 50.0
COMMERCIAL_FIXED_CHARGE = 150.0
INDUSTRIAL_FIXED_CHARGE = 300.0

# 3. Define a global constant for the customer charge
CUSTOMER_CHARGE = 20.0

# 4. Define a global constant for the electricity duty percentage
ELECTRICITY_DUTY_PERCENTAGE = 0.05

def calculate_energy_charges(units_consumed, customer_type):
    """
    Calculates energy charges based on units consumed and customer type.
    Returns energy charges and the rate applied per unit.
    """
```

Variables Terminal

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lab 3.3.ipynb

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Commands + Code + Text Run all

RAM Disk

```
[15] rate_applied = COMMERCIAL_RATE_PER_UNIT
elif customer_type == 'industrial':
    rate_applied = INDUSTRIAL_RATE_PER_UNIT
else:
    raise ValueError("Invalid customer type")
energy_charges = units_consumed * rate_applied
return energy_charges, rate_applied

def calculate_fixed_charges(customer_type):
    """
    Calculates fixed charges based on customer type.
    Returns the fixed charges.
    """
    fixed_charges = 0.0
    if customer_type == 'domestic':
        fixed_charges = DOMESTIC_FIXED_CHARGE
    elif customer_type == 'commercial':
        fixed_charges = COMMERCIAL_FIXED_CHARGE
    elif customer_type == 'industrial':
        fixed_charges = INDUSTRIAL_FIXED_CHARGE
    else:
        raise ValueError("Invalid customer type")
```

Variables Terminal

11:11 PM

Snipping Tool

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lab 3.3.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[15] ✓ 0s

```
def calculate_electricity_duty(energy_charges, fixed_charges, customer_charges):  
    """  
    Calculates electricity duty based on total charges.  
    Returns the electricity duty amount.  
    """  
    total_charges_before_duty = energy_charges + fixed_charges + customer_charges  
    electricity_duty = total_charges_before_duty * ELECTRICITY_DUTY_PERCENTAGE  
    return electricity_duty  
  
print("Constants and functions defined successfully.")
```

... Constants and functions defined successfully.

[16] ✓ 12s

```
print("\n--- Electricity Bill Calculator ---")  
  
# 1. Read inputs from the user  
try:  
    previous_units = int(input("Enter Previous Units: "))  
    current_units = int(input("Enter Current Units: "))  
    customer_type_input = input("Enter Customer Type (domestic, commercial, industrial): ").lower()
```

Snipping Tool

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lab 3.3.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[18] ✓ 0s

```
try:  
    # Calculate actual_units_consumed  
    actual_units_consumed = current_units - previous_units  
  
    # Handle cases where Current Units are less than Previous Units  
    if current_units < previous_units:  
        print("Warning: Current Units are less than Previous Units. Billing units will be set to 0.")  
        units_for_billing = 0  
    else:  
        units_for_billing = actual_units_consumed  
  
    # Validate customer type  
    if customer_type_input not in ['domestic', 'commercial', 'industrial']:  
        raise ValueError("Invalid customer type provided.")  
  
    # Call calculate_energy_charges  
    energy_charges, rate_applied = calculate_energy_charges(units_for_billing, customer_type_input)  
  
    # Call calculate_fixed_charges  
    fixed_charges = calculate_fixed_charges(customer_type_input)
```

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lab 3.3.ipynb

File Edit View Insert Runtime Tools Help

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RAM Disk

```
[18] ✓ Os # Call calculate_fixed_charges
fixed_charges = calculate_fixed_charges(customer_type_input)

# Call calculate_customer_charges
customer_charges = calculate_customer_charges()

# Call calculate_electricity_duty
electricity_duty = calculate_electricity_duty(energy_charges, fixed_charges, customer_charges)

# Calculate total_bill_amount
total_bill_amount = energy_charges + fixed_charges + customer_charges + electricity_duty

# Display Bill Summary
print("\n--- Electricity Bill Summary ---")
print(f"Previous Units: {previous_units} units")
print(f"Current Units: {current_units} units")
print(f"Customer Type: {customer_type_input.capitalize()}")
print(f"Actual Units Consumed: {actual_units_consumed} units")
print(f"Units Billed: {units_for_billing} units")
print(f"Rate Applied: ₹{rate_applied:.2f} per unit")
print("\n--- Charges Breakdown ---")
print(f"Energy Charges: ₹{energy_charges:.2f}")
```

Variables Terminal

11:11 PM Python 3

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https://colab.research.google.com/drive/1uaaMltcZ3gnrjdHa-zqUN3sjLqtEbRpi#scrollTo=49cd82ab

lab 3.3.ipynb

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Commands + Code + Text Run all

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```
[18] ✓ Os print(f"Customer Charges: ₹{customer_charges:.2f}")
print(f"Electricity Duty: ₹{electricity_duty:.2f}")
print("-----")
print(f"Final Total Bill Amount: ₹{total_bill_amount:.2f}")
print("-----")

except ValueError as e:
    print(f"Error: {e}")
except Exception as e:
    print(f"An unexpected error occurred: {e}")

# Sample Run 1: Domestic Customer
run_billing_simulation(100, 250, 'domestic')

# Sample Run 2: Commercial Customer
run_billing_simulation(200, 450, 'commercial')

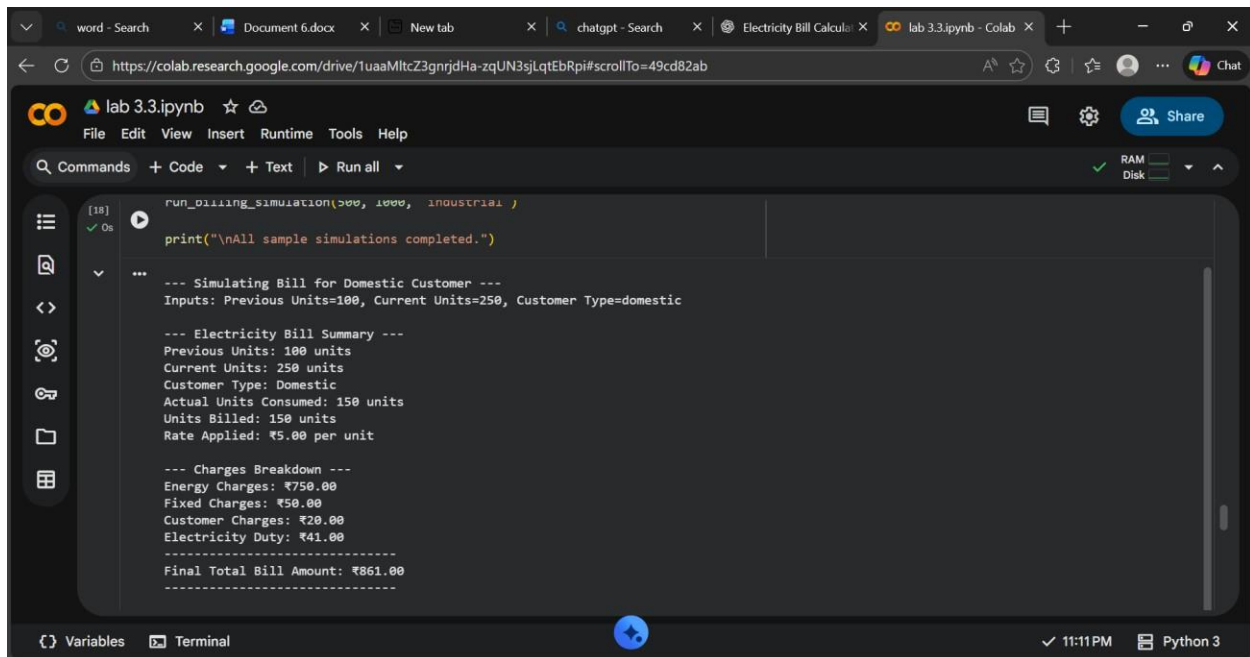
# Sample Run 3: Industrial Customer
run_billing_simulation(500, 1000, 'industrial')

print("\nAll sample simulations completed.")
```

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Variables Terminal

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```
[18] In [18]: run_billing_simulation(100, 250, 'domestic')
Out[18]: print("\nAll sample simulations completed.")

--- Simulating Bill for Domestic Customer ---
Inputs: Previous Units=100, Current Units=250, Customer Type=domestic

--- Electricity Bill Summary ---
Previous Units: 100 units
Current Units: 250 units
Customer Type: Domestic
Actual Units Consumed: 150 units
Units Billed: 150 units
Rate Applied: ₹5.00 per unit

--- Charges Breakdown ---
Energy Charges: ₹750.00
Fixed Charges: ₹50.00
Customer Charges: ₹20.00
Electricity Duty: ₹41.00
-----
Final Total Bill Amount: ₹861.00
-----
```

Output:

Explanation: The final program accurately computes all billing components and presents them in a clear, structured format.

Its readable logic and modular design make it suitable for real-world electricity billing applications and easy future enhancements.